Consolidated Findings: Simple Policies, Information, and the Room for RL in the Pharmaceutical Supply-Chain Simulator

One running document consolidating the routing study, the routing-repaired (“understanding”) study, and the value-decomposition study (Phases 1–2, distributor-seat test, Part C). Individual reports remain archived as the audit trail.

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How to read this document

The setting. Everything in this document comes from a simulated pharmaceutical supply chain inherited from a predecessor thesis: two manufacturers ship to two distributors, which ship to two health centers serving patient demand (a 2×2×2 network). In the central experiment, one manufacturer loses 95% of its production capacity for 48 of the episode’s 300 periods. The thesis trained a deep reinforcement-learning (RL) agent to control the distributors and reported that it far outperforms a standard base-stock (order-up-to-a-target) policy under this disruption. Our research question: *can simple, transparent decision rules match the thesis’s RL agent and beat the base-stock baseline — and if not, where does the remaining value for learning actually live?*

What this document is. The single source of current truth consolidating three experiment campaigns (the routing study, the routing-repaired “understanding” study, and the value-decomposition study). It is ordered most-load-bearing first: Section 1 lists every claim with its evidence and status; Section 2 gives the four findings that matter; Section 3 carries the phase-by-phase detail; Section 4 lists superseded claims (do not cite); the closing sections record the verification regime and next steps. The full individual reports remain archived as the audit trail.

Term used below	Meaning
thesis world ("as-shipped" routing)	The simulator exactly as the thesis configured it. In particular, one health center splits its orders between the two distributors by an internal trust score; the other is hard-wired to split 50/50 (the "captive" customer — it cannot shift orders away from a failed supplier).
routing-repaired world	The same simulator with our repair to the health centers' routing layer, so orders can follow surviving supply. Used to study what remains once the routing artifact is removed.
trust	The simulator's built-in supplier rating: a health center routes a larger share of its orders to suppliers that have delivered reliably in the recent past.
disruption cost	System-wide holding-plus-backlog cost accumulated from the disruption's onset to the end of the episode.
thesis severity (slack regime)	The disruption as the thesis configured it: one manufacturer at 5% capacity for 48 periods, the *surviving* manufacturer unconstrained — it has spare capacity and could serve all demand if orders were rerouted to it.
moderate / severe scarcity	Harder regimes in which the surviving plant is *also* cut, to 50% (moderate) or 30% (severe) of capacity, so total supply is below total demand and no routing rule can serve everyone.
urgent0 / urgent20	The two demand configurations: all patient demand deferrable (unmet demand is backlogged) vs. 20 urgent patients per health center per period whose demand is permanently lost if unmet that period.
premium / coverage	Insurance framing for standing buffers: the *premium* is the extra normal-times cost a policy pays; the *coverage* is the disruption damage it removes.
reporting seeds	20 held-out random demand scenarios (seeds 11–30) never used for tuning; every headline number is measured on these.
the compound	Shorthand for the named combination policy of the section at hand (components defined where used). The distributor-seat compound, e.g., is serve-the-captive routing × stop ordering from the dead factory × a standing buffer.
oracle	A non-deployable reference policy given perfect advance knowledge of the disruption onset; used as an upper bound, not a proposal.
base stock	The standard order-up-to-a-target inventory policy; the thesis's (and our) baseline.
MN / DS / HC	Manufacturer / distributor (the seat the RL agent controls) / health center.

1 Current state of claims

Every load-bearing claim, written as a self-contained sentence, grouped into four themes. All values come from audited runs on held-out reporting seeds unless marked otherwise. *Status* codes: **established** (audited and standing), **corrected** (replaces an earlier number or reading — the superseded version is listed in Section 4), **open** (cannot yet be verified), **retracted** (withdrawn; do not cite).

1.1 The disruption and the routing layer

Claim (plain language)	Evidence (key numbers)	Status
The cost pile-up that follows the manufacturer outage is mostly an artifact of the simulator's routing layer (the captive health center's hard-wired equal split, and the fact that orders stranded in a dead supplier's queue are never re-issued), not unavoidable supply physics: simple health-center routing rules alone remove most of the disruption cost at thesis severity.	67–91% of the disruption cost is removable by HC routing rules alone (thesis severity).	established
Downstream detection of the outage is structurally blind for about six periods: the failed distributor keeps shipping from its own stock, and that stock cover masks the failure from every downstream observer. An explicit delivery-rate detector performs *worse* than the simulator's built-in trust mechanism.	$\sim$6-period blind window; $\sim$0.9M of cost accrues behind the stock-cover mask; explicit detector $<$ built-in trust.	established
The earlier claim that "detection captures approximately zero" was too broad and has been narrowed: detection followed by *buffering* is indeed worth approximately zero, but detection followed by *rerouting* carries the value of knowing the onset.	detect-then-buffer ≈ 0; detect-then-reroute carries the onset value.	corrected

1.2 Information and foresight

Claim (plain language)	Evidence (key numbers)	Status
Sharing the upstream machine-state signal is worth exactly as much as perfect knowledge of the disruption onset: a two-line reroute triggered by the shared signal reproduces the perfect-onset oracle's trajectory bit-for-bit.	Bit-identical trajectories (audited); disruption cost 1.06M $\rightarrow$ 295k; lost patients 445 $\rightarrow$ 203.	established
True foresight — acting *before* the onset — adds a small but real margin on top of the shared signal, provided the preparatory buffer is placed at the correct location (the surviving chain). This corrects our earlier claim that foresight adds nothing beyond the shared signal, which was an artifact of a mis-located buffer.	$\approx$54k ($\sim$5%) of additional value with a correctly located buffer.	corrected
In the slack regime, the remaining room for *any* policy — learned or not — above the best deployable compound is approximately zero *within the enumerated rule family.* This is a family-bounded statement, not an absolute one, though no candidate mechanism outside the family has been identified.	Room above best deployable ≈ 0 (family bound; moderate-high confidence).	established

1.3 The distributor's levers (the RL question)

Claim (plain language)	Evidence (key numbers)	Status
The best simple distributor compound — serve-the-captive routing × stop-ordering-from-the-dead-factory × a standing buffer — removes about half of base stock's profit loss in the thesis's own world, on the thesis's own metric, in both demand configurations (all demand deferrable, "urgent0"; 20 urgent patients per health center per period, "urgent20").	49.1% (urgent0) / 50.7% (urgent20) of base stock's loss removed; fill 0.858 vs 0.808; lost patients 1,575 vs 1,768.	established
The thesis RL result (Table 3.9) claims to remove 89% of base stock's loss, but it is UNREPLICATED and the baseline configuration is unknown; we treat it as a claim, not a reproduced result, pending the author's configuration.	Claimed 89% vs our measured 49–51% for simple rules; baseline config unknown.	open
Demand-shaping from the distributor's seat is real: deliberately shedding one health center's demand improves the system on episode totals, and the *direction* of the shed matters (shedding the wrong customer destroys value). This supersedes the earlier reading that allocation is worth ≤12% everywhere, which was specific to the routing-repaired world.	shed +21.8% (both HCs better off on totals); direction control: shed_inverse −14.3%.	established
Tapering orders toward the failed source looks valuable on the thesis metric, but the gain is accounting-driven (dead-factory queue bookkeeping), and the lever is harmful to patients when used alone.	+24.9% on the thesis metric; +9% lost patients when used alone.	established
A state-elevated base-stock policy is dead on arrival in this simulator: supply physics prevent inventory from climbing during the outage, so raising the target does nothing.	Inventory cannot climb during the outage (4 units measured).	established
In the routing-repaired worlds, the distributor's allocation channel is bounded at ≤12% (static priority, costed for fairness at ±0.023 fill); every dynamic "smart" allocation rule backfires through the trust loop, and static priority itself flips harmful under scarcity.	≤12% bound; dynamic rules backfire +25–56%; priority +23% harmful under scarcity.	established
Our repaired DRL pipeline (six reward components fixed, exploration schedule linearized) trains stably but *loses* to base stock on system profit; there is currently no working RL comparator in this codebase.	System profit −4.06M (DRL) vs −2.35M (base stock).	established

1.4 Scarcity: buffers and the frontier

Claim (plain language)	Evidence (key numbers)	Status
Under moderate scarcity (the surviving plant also cut to 50%, 48-period outage), the frontier is owned by one composition: the compound of buffer-at-survivor, reroute, and taper removes most of the cost at near-perfect fill, and dominates buffer-only at the same buffer premium.	−79% of cost at fill 0.99–1.00; dominates buffer-only 2.4× at matched premium.	established
The buffer's optimal *location* flips to the disrupted chain when scarcity is severe enough that the surviving chain cannot carry the load. The lever-flip map orders the winning lever across the severity × duration plane: no-action (duration ≤ stock cover) → routing → compound → buffer-only.	Severe-scarcity cell corrected: disrupted-located buffer wins at 2,417k, −30% vs the healthy-chain location.	corrected (severe-scarcity cell)
Demand-aware buffer *sizing* — set $B = (\text{demand} - \text{surviving deliverable}) \times \text{per-event duration}$ — replaces the frozen per-severity grids and dramatically cuts lost patients. The per-event qualifier comes from the recurrence check: buffers refill between events, so sizing to the cumulative duration over-buys.	Lost patients 505 → 41 at the frozen optimum; under two events, per-event sizing beats total-duration sizing (1,033k vs 1,210k).	established (refined)
Severity-aware redirect caps are REFUTED in both tested forms: capping redirected orders starves the surviving supplier of the order signal that drives its production, so the cap hurts exactly when it is meant to help.	Both forms refuted; mechanism: orders are the supplier's production signal.	established (negative result)
Buffers in the slack regime are location-specific: a buffer at the disrupted chain is worth zero-to-negative, while a buffer at the surviving chain adds a few percent. This corrects both the blanket "slack buffers are negative" claim and the broken-routing "insurance captures 42%" artifact.	Disrupted chain: zero-to-negative; surviving chain: +4–7%.	corrected

1.5 Pre-publication robustness checks (all passed)

Claim (plain language)	Evidence (key numbers)	Status
The lever-flip map is robust to recurring disruptions: with two identical events per episode, no cell changes its winning lever, and the compound's advantage *grows* under recurrence — its signal-gated components cost nothing between events and its buffer refills, so its cost grows only ∼1.35–1.4× for two events. The no-action zone survives recurring short blips, and aggressive rerouting remains the worst policy at severe scarcity.	Compound, two events: 378k (mild) / 1,033k (moderate); no-action 142,639 still beats routing 203,544 for recurring blips; routing 10.1M still worst at severe scarcity.	established
Coverage does not collapse in normal times: in episodes with NO disruption, the signal-gated rules are inert (within 0.5% of base stock's profit), so the information-sharing compound pays essentially zero premium — in sharp contrast to the thesis RL, which lost 18% in its own no-disruption scenario. Standing buffers, by contrast, pay their full holding premium when nothing happens.	No-disruption profit: base stock 794,889; gated compound 790,732 (−0.5%); compound with standing buffer 527,072 (−34%).	established
Equity of the shed rule: on episode totals BOTH health centers are better off (fill and lost patients improve for both), but the transition has a real, localized cost — the deliberately starved health center's worst 5-period service dips from 0.614 to ≈0.50 before the rerouting completes. Honest deployment framing: Pareto-improving on totals, with a measurably deeper short trough for the starved customer; pair with a service floor in practice.	Lost patients per HC (urgent20): trust-routed 804 → 713; captive 964 → 862; worst 5-period fill 0.614 → ≈0.50.	established

2 The four findings that matter

F1 — If the distributors can see the manufacturers' machine status, a two-line rule is as good as perfectly predicting the disruption — and without that visibility, no algorithm of any sophistication can react in time. The evidence: at thesis severity the surviving manufacturer has 3.3× headroom and ramps its production with the orders it receives; a two-line reroute triggered by the shared upstream machine-state signal exactly reproduces the perfect-onset oracle (bit-identical trajectories). Without that sharing, no downstream policy — learned or not — can act in time, because the failed distributor's own stock masks the failure from every downstream observer for about six periods. *Meaning: the thesis's information-sharing narrative is right, but its value is captured by a trivial rule, not by a learner.*

F2 — A three-parameter rule recovers about half of the improvement the thesis's RL agent claimed over base stock; whether the other half is real cannot be judged until the thesis's baseline configuration is found. The evidence: from the distributor's seat, in the thesis's own world and on the thesis's own metric, the compound (serve-the-captive routing × a taper that stops ordering from the dead factory × a standing buffer) removes 49–51% of base stock's loss — against the RL's claimed 89% from a thesis table we cannot replicate (our own corrected-pipeline DRL agent *lost* to base stock). *Meaning: simple rules beat base stock decisively; whether the RL's remaining margin is real is now a question about the thesis baseline configuration — the single most valuable item to obtain.*

Equity of the shed rule: on episode totals both health centers are better off under the shed compound (with urgent patients present — the urgent20 configuration — lost patients fall at the trust-routed health center 804→713 and at the captive health center 964→862; during-disruption fill rises for both), but the transition has a real localized cost: the deliberately starved health center's

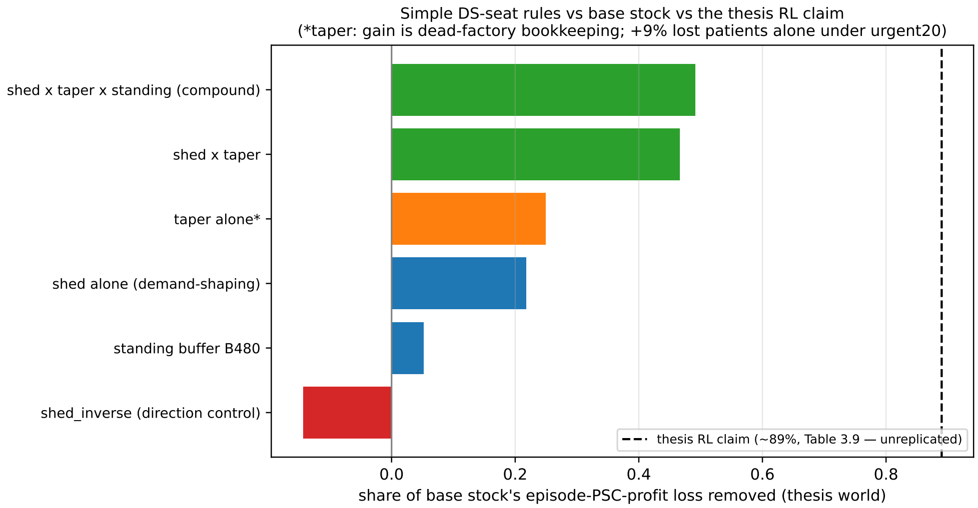

Figure 1: F2 in one picture: simple distributor rules vs the RL claim, thesis world, audited reporting seeds.

worst 5-period fill deepens from 0.614 (base stock) to ≈0.50 before the reroute completes. The rule is Pareto-improving on totals with a measurably deeper short trough for the starved customer — the honest ethics framing for any deployment.

F3 — When supply is genuinely short, what matters is where reserve stock sits and how big it is — both computable in advance from observable quantities; reacting cleverly during the outage does not help. The evidence: under scarcity the cost–service frontier is owned by a single composition (buffer + reroute + taper); the buffer's optimal *location* flips to the disrupted chain once the surviving chain is too weak to carry the load; and demand-aware *sizing* — buffer = (demand − surviving deliverable) × duration — cuts lost patients from 505 to 41. *Meaning: the two deployable decisions that matter are where stock sits and how it is sized — both computable from observables, no learning required.*

F4 — Every adaptive "smart" rule we tested made the system worse; simple static structure won every time. The evidence: across three studies, every reactive rule backfired — delivery-rate detection, backlog-priority allocation, serve-the-captive (in the routing-repaired world), rotating priority, priority-with-floor, and both severity-aware redirect caps. The mechanism is structural: the trust feedback loop rewards consistency, and the surviving manufacturer's production is driven by the orders it receives, so clever order withholding starves the very supply it is trying to protect. *Meaning: for the eventual RL comparison, the honest prior is that adaptivity must overcome a measured pattern of adaptive rules underperforming static structure.*

3 Phase-by-phase detail

3.1 Background: the DRL arc (why there is no working RL comparator)

The reward audit found reward components 1 and 4 carrying a thesis-level algebraic defect, component 5 a code defect, and components 2/3/6 saturated as configured; repairs un-stuck all six components. Training instability traced to the adaptive exploration rule (a multiplicative update that runs away to its cap); a linear schedule yields a stably-training agent — which then *loses* to base stock on system profit (−4.06M vs −2.35M, driven by manufacturer queue cost). The thesis's

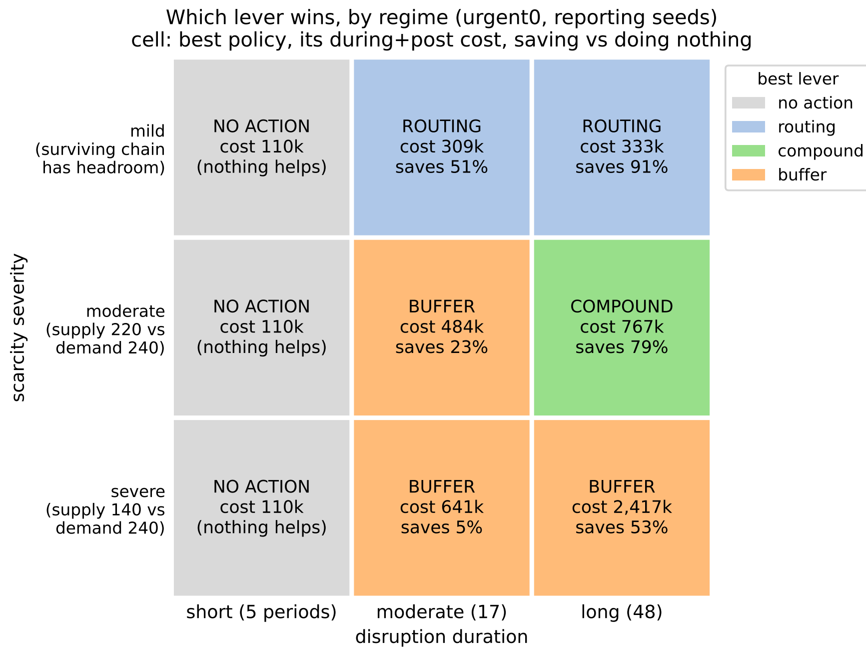

Figure 2: F3 in one picture: the lever-flip map over the severity × duration plane. Each cell's color indicates the winning lever (no-action → routing → compound → buffer-only); the severe-scarcity cell is corrected to a disrupted-located buffer. Cell color encodes WHICH lever wins; cell text gives that policy's cost and its saving versus doing nothing. The recurring-disruption robustness check has since been run and PASSED: with two events per episode no cell changes its winner, so the map stands.

headline RL result (its Table 3.9) is therefore treated as a claim, not a reproduced result, pending the author's configuration.

3.2 Routing study (thesis severity)

The captive health center's hard-wired equal split, plus stranded-order suppression (orders stuck in a dead supplier's queue are never re-issued), lock ∼2/3 of the disruption cost into the routing layer: trust-split −26%, sharpened trust + stale-order write-off −67% (fill 1.000), oracle onset reroute −91%; lost patients 1,768 → 203. Found en route: the normal demand model hard-reseeds the global random number generator (all historical multi-seed normal-demand results were a single demand path) — fixed in our runners, gate-guarded.

3.3 Understanding study (routing-repaired world)

Allocation bounded at ≤12% via the maximum over the enumerated rule family (static priority; fairness cost ±0.023 fill); all dynamic rules backfire through the trust loop. The residual disruption cost in the routing-repaired world decomposes into 41% dead-factory queue bookkeeping (dual cost accounting was introduced for exactly this), the ∼10-period redirection transient, and a write-off glut later removed as a side effect of instant rerouting.

3.4 Value decomposition, Phase 1 (slack regime)

Info-sharing reroute ≡ oracle (bit-identical, audited); explicit delivery-rate detection worse than the built-in trust mechanism (stock-cover masking ∼0.9M); deployable compound 273k. Verification rebuilt the oracle as a true superset and corrected two claims: foresight is worth ∼54k with the buffer at the SURVIVING chain, and slack-regime buffers are negative only at the disrupted location (+4–7% at the survivor).

3.5 Value decomposition, Phase 2 (scarcity regimes)

With the surviving manufacturer also cut (to 50% / 30%), routing still reduces cost (−26%/−50%) but cannot create product (lost patients ∼flat across the entire ladder); the compound reaches −79% at fill 0.99–1.00; static priority flips harmful (+23%); the severity×duration map has two boundaries (stock-cover duration; surviving-chain headroom), with sharp rerouting the WORST measured policy at severe scarcity.

3.6 Distributor-seat study (thesis world) and Part C

Full-grid ladder over the distributor's complete decision space (everything else is dead by topology). Results in F2/F4 and the claims tables. Part C: demand-aware buffer sizing (the robust knob), the severe-scarcity buffer-location correction, and the redirect-cap refutation with its mechanism (orders are the surviving supplier's production signal).

4 Superseded claims (do not cite)

- The −468k / −654k value split — measured on the broken routing layer; retracted.
- "Permanent insurance captures 42%" — broken-routing artifact; in the routing-repaired slack-regime world, disrupted-located buffers are zero-to-negative (surviving-located: +4–7%).
- "Foresight adds nothing beyond the shared signal" — artifact of a mis-located ceiling buffer; corrected to ∼54k (∼5%).
- "Detection captures ≈0" — true only for detect-then-buffer; rerouting responses carry most of the onset value, and the upstream-signal trigger captures it fully.
- "Allocation ≤12% is the whole channel" — specific to the routing-repaired world; in the thesis world the demand-shaping channel is worth +21.8% (shed) and the compound 49%.
- "Fill 1.000" for the 960-unit-buffer scarcity compound — display-rounding; 0.990 (the 1,440-unit buffer reaches 0.9999).
- The frozen per-severity buffer grids — replaced by demand-aware sizing.

5 Rigor record (one paragraph)

All experiments ran on a gated harness (bit-exact baseline reproduction on both library and CLI paths, determinism, demand conservation, cross-seed variance), with pre-registered tuning/reporting seed splits (1–10 / 11–30), dual cost accounting, and both fairness baselines carried. Every phase was audited by an independent agent re-deriving headline numbers from raw CSVs (12/12, 11/12 — the exception a display-rounding over-claim, corrected — 16/16, 13/13, and 18/18 for the robustness checks). Three contaminated or buggy batches were caught by gates or diagnostics (CLI-default contamination; a bootstrap deadlock; both logged) and discarded without contaminating conclusions. Simulator defects found and guarded: the normal demand model's global RNG reseed, a dead trust-delta config constant, the silent discard of orders $\leq$1 unit, and the missing on-order timeout that produces stranded-pipeline suppression.

6 Next steps

1. **Obtain the thesis Table 3.9 configuration (incl. the base-stock baseline)** — decides the interpretation of F2's 49%-vs-89% gap; everything else is measured.
2. *(Done since the last draft:)* the recurring-disruption robustness check PASSED — the map's winners are unchanged and the compound amortizes best; coverage does not collapse in normal times (gated rules pay $\approx$zero premium); the shed rule's equity profile is characterized. The map can be published as-is.
3. Ramped-onset regime as the second paper (the one place detection could earn value).
4. When corrected DRL code arrives: evaluate against the regime-appropriate simple compounds (thesis world: the distributor-seat compound; slack regime: 273k; moderate scarcity: 767k–1,726k with demand-aware buffer sizing), same seeds, same gates, both accounting conventions.
5. Distill the paper from this document: the three-lever value decomposition with the lever-flip map, the distributor-seat result as the RL-relevant centerpiece.